ISBAT UNIVERSITY

Faculty of Information and Communication Technology (FICT)

REQUIREMENTS GATHERING & REQUIREMENTS ENGINEERING DOCUMENT

for the

Association Boda-Boda Ride Booking and Dispatch Application

Prepared as a Reference Document for: Software Engineering and Professional

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Document Version: 1.0

Phase: Requirements Engineering (precedes the SRS)

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1. Introduction

1.1 Purpose of this Document

This document records the Requirements Engineering (RE) process followed for the **Boda-Boda Ride Booking & Dispatch App** a proposed system that will allow passengers to request boda-boda rides, receive estimated fares, be matched with available verified riders, track their rides in real time, and pay through mobile money or cash. The system will be owned and operated by the boda-boda association, with association management and dispatch/control room staff responsible for monitoring rides, managing riders, and supporting the dispatch process.

Unlike a Software Requirements Specification (SRS), which records the final agreed requirements, this document captures **HOW** those requirements were discovered, analysed, validated, and will be managed, the requirements engineering process and artefacts produced before and alongside the SRS.

1.2 Project Background

Boda-boda transportation in Kampala is currently handled largely through informal, manual processes, where passengers identify and contact nearby riders directly, negotiate or agree on fares, and coordinate pickup locations through phone calls or physical interaction. This can result in difficulties finding available riders, inconsistent fare pricing, limited visibility of ride progress, and safety concerns due to limited rider verification and tracking. The boda-boda association has proposed the **Boda-Boda Ride Booking & Dispatch App** to digitise the ride-booking and dispatch workflow, providing passengers with a convenient way to request rides, receive estimated fares, make payments through mobile money or cash, and track their rides, while enabling verified riders and association dispatch staff to manage and monitor ride requests efficiently.

1.3 Objectives of the Requirements Engineering Phase

- Identify and engage all stakeholders affected by, or with influence over, the boda-boda ride booking and dispatch process, including passengers, boda-boda riders, association management, dispatch/control room staff, and mobile money service providers.
- Elicit a complete, accurate, and prioritised set of requirements for ride booking, rider dispatch, fare estimation, payment, real-time location tracking, notifications, and safety using appropriate requirements elicitation techniques.
- Analyse, classify, and resolve conflicts among stakeholder requirements, particularly where passenger convenience, rider needs, operational efficiency, safety, and association policies may conflict.
- Validate requirements with stakeholders before system design and development begins, ensuring that the proposed booking, dispatch, payment, and safety processes accurately reflect the needs of the association and its users.
- Establish a requirements management process covering change control, requirements traceability, prioritisation, and versioning throughout the development and implementation of the system.

1.4 Relationship to the SRS

This document feeds directly into the **Software Requirements Specification (SRS)** for the Boda-Boda Ride Booking & Dispatch App. Requirements gathered, analysed, and validated during the Requirements Engineering process are refined into formal **Functional Requirements (FRs)** and **Non-Functional Requirements (NFRs)** in the SRS. The RE process is iterative: elicitation, analysis, and validation activities may be repeated as the association, passengers, riders, and other stakeholders develop a deeper understanding of the ride booking, dispatch, payment, and safety requirements.

2. Stakeholder Identification and Analysis

Correctly identifying stakeholders is the first and most critical step in requirements engineering — missed stakeholders lead to missed requirements. For the **Boda-Boda Ride Booking & Dispatch App**, stakeholders are identified based on their involvement in the ride-booking and dispatch process, their interest in the system, and their influence over its success. The stakeholders are classified according to their level of **power/influence** and **interest** in the project, following the classic **Power–Interest Grid**.

Stakeholder	Role / Interest	Power	Interest	Engagement Strategy
Passengers	Primary end-users; request rides, view estimated fares, track rides, and make payments.	Medium	High	Keep informed; gather requirements through surveys, interviews, and usability testing.
Boda-Boda Riders	Accept ride requests, navigate to pickup locations, transport passengers, and receive payments.	Medium	High	Consult closely; involve riders in requirements elicitation, workflow testing, and pilot deployment.
Boda-Boda Association Management	Owns and manages the system; sets operational policies, manages riders, monitors performance, and oversees the service.	High	High	**Manage closely — key decision-makers;** involve them in requirements approval and major system decisions.

Dispatch / Control Room Staff	Monitor ride requests, assign or oversee dispatch, handle exceptions, and support passengers and riders during rides.	High	High	**Manage closely —** involve them in dispatch workflow design, testing, and operational validation.
Mobile Money Provider	Provides the mobile money payment infrastructure/API for processing ride payments and confirming transactions.	Medium	Medium	Keep satisfied; involve technical representatives during payment integration and testing.
Association IT / System Administrators	Manage system infrastructure, user accounts, security, backups, and technical maintenance.	High	High	**Manage closely technical stakeholders;** involve them in architecture, security, integration, and maintenance requirements.
Boda-Boda Association Members / Owners	Have an interest in rider registration, compliance, revenue collection, and the overall effectiveness of the association's service.	Medium	High	Keep informed; consult them on operational policies, reporting, and rider management requirements.
Association Top Management / Sponsors	Approve the project, provide funding, establish strategic direction, and approve major operational changes.	High	Medium	Manage closely; provide periodic progress reports and seek approval at major project milestones.

2.1 Stakeholder Conflicts to Watch

- Association management wants strict rider verification and registration requirements; riders may want a faster and less restrictive onboarding process — to be resolved through requirements negotiation, with the final verification criteria captured as clear business rules.

- Passengers want accurate and affordable fares; riders want fares that adequately compensate them for distance, time, and operating costs — requires agreement on the fare calculation model and pricing rules rather than leaving the decision to the technical team.
- Dispatch/control room staff want authority to manually assign or reassign rides; riders may prefer to accept rides directly based on their availability and location — requires a clear dispatch policy defining when automatic dispatch, rider acceptance, and manual intervention should apply.
- Passengers want real-time rider location and trip tracking for safety; riders may have privacy concerns about continuous location monitoring — requires a defined location-sharing policy specifying when location data is collected, who can access it, and how long it is retained.
- Passengers may prefer cash payments; the association may prefer mobile money because it provides easier transaction tracking and reconciliation — requires a business policy defining supported payment methods and how cash transactions will be recorded and reconciled.

3. Requirements Elicitation Techniques

A combination of elicitation techniques was used to gather a rounded picture of requirements from different stakeholder groups, since no single technique surfaces all types of requirement.

3.1 Techniques Applied

Technique	How It Was Applied on OCRS	Best For
One-on-one Interviews	Structured interviews with association management, dispatch/control room staff, riders, and selected passengers to understand operational rules, safety concerns, dispatch processes, and pain points.	Deep, detailed requirements from key decision-makers and operational stakeholders.
Questionnaires / Surveys	An online or mobile survey distributed to a sample of passengers and boda-boda riders to gather input on booking preferences, fare expectations, payment methods, safety concerns, and usability.	Gathering input from a large number of users.
Focus Group Discussion	A facilitated session with passengers and riders to explore common experiences, frustrations, expectations, and preferences regarding ride booking, fares, safety, and communication.	Surfacing requirements through group interaction and debate.
Document / Process Analysis	Review of existing association records, rider registration procedures, fare guidelines, dispatch procedures, payment records, and safety policies.	Understanding existing business rules, processes, and terminology.
Direct Observation	Observing boda-boda operations at selected stages, including how passengers request rides, riders are assigned, pickup locations are coordinated, fares are agreed, and rides are monitored.	Discovering workarounds, informal practices, and implicit requirements that stakeholders may not mention.
JAD(Joint Application Development) Workshop	A facilitated workshop involving association management, riders, dispatch staff, and passenger representatives to agree on the end-to-end ride booking, dispatch, payment, and safety workflow.	Resolving stakeholder conflicts and reaching consensus quickly.
Low-Fidelity Prototyping	Paper sketches or simple wireframes of the passenger booking screen, fare estimate, rider acceptance screen, live ride tracking, and payment confirmation screens are tested with representative users.	Validating usability assumptions and user-interface requirements before development.

3.2 Elicitation Schedule

Activity	Participants	Duration	Output
Association Management Interview	Association Chairperson / Manager	60 minutes	Interview notes; association policies, operational rules, and business requirements
Dispatch Staff Interview	Dispatch/control room staff	60 minutes	Dispatch procedures, ride assignment rules, escalation and incident-handling requirements
Rider Interviews	5–10 registered boda-boda riders	60 minutes	Rider requirements; acceptance, navigation, fare, payment, and availability requirements
Passenger Survey	100 passengers (stratified sample)	2-week open window	Quantitative data on booking preferences, fare expectations, payment methods, and safety concerns
Passenger Focus Group	8 passengers	90 minutes	Qualitative pain-point themes and user expectations
Process Observation	Riders, dispatch staff, selected pickup locations	3 mornings	Observation log of the current ride-request, rider-assignment, pickup, and payment workflow
JAD Workshop	Association management, dispatch staff, riders, passenger representatives, IT representative	Half day	Agreed end-to-end ride booking, dispatch, payment, and safety workflow
Prototype Walkthrough	5 passengers and 5 riders	30 minutes each	Usability feedback log; validated interface and workflow requirements

4. Elicitation Instruments

4.1 Sample Interview Guide — Boda-Boda Association Management

1. Walk me through how a passenger currently requests and completes a boda-boda ride, from the initial request to payment and drop-off.
2. What information must be verified before a rider is allowed to accept passenger requests (e.g., rider registration, identification, motorcycle details, licence, or association membership)?
3. How are riders currently assigned to passengers, and what factors should the system consider when dispatching a ride (e.g., distance, rider availability, location, or passenger preferences)?
4. How are fares currently determined, and what rules should the system use to calculate and display an estimated fare before a passenger confirms a ride?
5. What reports and information does the association management need to monitor the service (e.g., completed rides, revenue, active riders, cancelled rides, and customer complaints)?
6. What is the single biggest cause of errors, delays, safety concerns, or customer dissatisfaction in the current ride-booking and dispatch process?

4.2 Sample Interview Guide — Mobile Money / Payment Provider

1. How are mobile money payments currently processed for services such as transportation, and what payment integration options does your platform provide?
2. What payment methods and transaction types should the ride-booking system support, and what requirements apply to each method?
3. How can a completed payment be linked to a specific passenger, ride, and transaction, and how is payment confirmation communicated to the system?
4. How are failed, cancelled, reversed, or refunded transactions handled, and what rules should the system follow when a payment does not complete successfully?
5. What transaction records, reconciliation information, and financial reports can the system generate, and which association staff should have access to them?

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4.3 Sample Passenger Questionnaire (Extract)

- On a scale of 1–5, how satisfied are you with the current process of finding and booking a boda-boda ride?
- How do you usually pay for boda-boda rides? **(Mobile Money / Cash / Other)**
- Have you ever experienced difficulty finding an available rider or coordinating a pickup location? **(Yes/No — if yes, describe briefly)**
- Would you prefer to request a ride using a **mobile phone app, USSD, phone call, or a combination of these?**

- How important is it for you to see the **estimated fare before confirming a ride? (Not important / Somewhat important / Very important)**
- How important is it for you to see the **rider's name, photograph, motorcycle details, and verification status** before starting a ride? **(Not important / Somewhat important / Very important)**
- Would you prefer to **share your live trip location with a trusted contact** while using the service? **(Yes / No / Not sure)**
- What single improvement would most improve your experience when booking a boda-boda ride?

5. Requirements Gathered (Raw, Pre-Analysis)

The list below captures raw statements as expressed by stakeholders during elicitation, before analysis, restructuring, or conflict resolution. Raw requirements are intentionally kept close to stakeholders' own words at this stage. For the **Boda-Boda Ride Booking & Dispatch App**, these statements represent the initial needs and expectations gathered from passengers, riders, association management, dispatch/control room staff, and the mobile money provider.

#	Source	Raw Requirement (stakeholder's words)	Type (initial)
R1	Association Management	"Riders should not be able to go online or accept ride requests if their stage registration or driver verification is incomplete."	Functional
R2	Finance / Dispatch	"We need to receive instant payment confirmations when a passenger pays, not wait until end-of-day reconciliation."	Non-functional (performance)

R3	Passenger survey	"I want to pay using MTN MoMo or Airtel Money directly inside the app without having to switch to a USSD code."	Functional
R4	Focus group (Passengers)	"The app should tell me exactly why my ride request failed, such as 'No riders in range', not just display a generic 'Error'."	Non-functional (usability)
R5	Boda-boda Rider	"The system must allow passengers to choose between cash or mobile money payment before requesting a trip."	Functional
R6	Technical Lead	"Whatever backend we build must run on low-cost cloud infrastructure to fit within the association's fixed operating budget."	Constraint
R7	Observation	"Passengers manually negotiate fares at physical stages; the app needs to automatically calculate and display an estimated fare before booking."	Functional
R8	Control room staff	"I need to see active riders and active trips update on the live dispatch map in real time."	Functional
R9	Association Management	"The app must not expose passenger contact numbers or rider identity documents to unauthorized third parties."	Non-functional (security)
R10	Dispatch Staff	"During Kampala evening rush hour, the system must handle hundreds of simultaneous ride bookings without crashing or freezing."	

6. Requirements Analysis and Classification

6.1 Analysis Process

Raw requirements were reviewed to remove duplication, resolve ambiguity, and separate genuine requirements from design suggestions or business rules. Each was reworded into a testable "the system shall..." statement and classified as Functional, Non-Functional, or Constraint.

6.2 Refined Requirements (Sample)

Ref	Refined Requirement	Type	Priority (MoSCoW)
R1	The system shall prevent a rider from operating if they haven't fully completed their registration with a national ID.	Functional	Must
R2	The system shall update a trip's payment status within 60 seconds of a confirmed mobile money transact:/ion.	Non-functional	Must
R3	The system shall allow a passenger to pay for a ride via MTN MoMo or Airtel Money directly within the application.	Functional	Must
R4	The system shall display a specific, human-readable reason whenever a ride booking attempt fails (e.g., "No nearby riders available").	Non-functional	Should
R5	The system shall allow a passenger to choose between cash or mobile money payment before confirming a trip booking.	Functional	Must
RG	The system shall be deployable on low-cost cloud server infrastructure without exceeding the association's monthly operational budget.	Constraint	Must
R7	The system shall allow a passenger to view and download a digital trip summary/receipt upon completion of a ride.	Functional	Should
R8	The system shall update the control room dispatch map in real time as riders change location or status.	Functional	Should
R9	The system shall restrict access to passenger contact details and rider verification documents to authorized Association Management and Dispatch staff only, based on role.	Non-functional	Must
R10	The system shall support at least 2,000 concurrent active ride request and tracking sessions during peak Kampala rush hours without performance degradation.	Non-functional	Must

6.3 MoSCoW Prioritisation Key

- Must have — Non-negotiable requirements critical for safe and functional go-live (e.g., rider verification, real-time dispatch, fare estimation, cash/mobile money payments). The app cannot launch without these.
- Should have — High-value requirements (e.g., digital ride receipts, live control room map updates) that are important, but the app can launch without them and add them in an immediate update post-launch.
- Could have — Desirable features (e.g., rider tipping, scheduled future bookings) that will only be included if time, budget, and development resources permit.
- Won't have (this release) — Features explicitly out of scope for Version 1.0 (e.g., inter-city parcel cargo tracking, multi-passenger ride pooling), which will be revisited for future system versions.

6.4 Conflict Resolution Log

Conflict	Resolution
Association Management wanted compulsory mobile money prepayments for all rides to eliminate cash theft; riders and passengers insisted on maintaining cash options due to network instability and user preference.	Resolved in the stakeholder workshop: passengers can choose either cash or mobile money upfront (R5), with automated instant payment updates configured for mobile money transactions (R2).
Association Management requested mandatory, continuous real-time GPS streaming for all registered riders; Technical Lead raised concerns over rider phone data consumption and server load during peak hours.	Real-time tracking retained during active trips and online dispatch mode (R8), with optimized location-ping intervals and a concurrency load-testing requirement (R10) added to manage peak Kampala rush hour traffic.

7. Requirements Validation

Validation confirms that the documented requirements are correct, complete, unambiguous, and reflect what stakeholders actually need — before design and development begin, when changes are cheapest to make.

7.1 Validation Techniques Used

- Requirements Walkthrough — The refined requirements list (Section 6.2) was reviewed line-by-line with Association Management, Stage Representatives, and Dispatch Operations staff to confirm operational accuracy and business alignment.

- Prototype Review — Interactive low-fidelity mock-ups of the passenger ride-booking and rider dispatch screens were tested with 5 boda-boda riders and 5 passengers; feedback was directly used to refine usability and system feedback requirements (R4).
- Checklist-Based Review — Each requirement was systematically checked against quality, clarity, and technical feasibility criteria (Section 7.2).
- Formal Sign-off — The Association Chairman, Head of Dispatch Operations, and Technical Lead signed off on the finalized requirements list (Section 9).

7.2 Validation Checklist

Quality Attribute	Validation Question
Correct	Does the requirement accurately reflect what the stakeholder actually needs?
Complete	Is anything missing that would prevent the system from being usable end-to-end?
Unambiguous	Could two different developers interpret this requirement in different ways?
Consistent	Does this requirement contradict any other requirement in the list?
Verifiable / Testable	Can a tester objectively confirm whether this requirement has been met?
Feasible	Can this be delivered within the project's budget, timeline, and infrastructure constraints (e.g., R6)?
Traceable	Can this requirement be traced back to a specific stakeholder need and forward to a design element and test case?

8. Requirements Management Plan

8.1 Requirements Traceability Matrix (Extract)

The traceability matrix links each requirement to its stakeholder source, the design element that will satisfy it, and the test case that will verify it. It is maintained for the life of the project and updated whenever a requirement changes.

Ref	Source	Requirement (short)	Design Element	Test Case ID
R1	Association Management	Rider verification & stage check	Rider Onboarding & Verification Module	TC-RIDE-01
R3	Passengers	In-app MoMo payment	Mobile Money Gateway Integration	TC-PAY-01
R5	Boda-boda Riders	Cash or mobile money payment selection	Payment Option & Fare Engine	TC-PAY-05
R9	Association Management	Role-based access to user/rider data	Authentication & Access Control Module	TC-SEC-02
R10	Dispatch Staff	2,000 concurrent sessions during rush hour	Load Balancer & Auto-Scaling Configuration	TC-PERF-01

8.2 Change Control Process

1. Any stakeholder (e.g., riders, passengers, dispatch staff, or association management) may submit a Requirement Change Request (RCR) describing the proposed change and its operational rationale.
2. The Business Analyst / Project Lead logs the RCR and performs an initial impact assessment (evaluating affected dispatch workflows, safety/compliance rules, technical design, project schedule, and operational costs).
3. The Change Control Board (Association Chairman/Management Representative, Head of Dispatch Operations, Technical Lead, and Project Manager) reviews and approves, rejects, or defers the change.
4. Approved changes update the system requirements list and traceability matrix, and the Software Requirements Specification (SRS) version number is incremented.
5. All relevant stakeholders are notified of approved changes before the next development iteration or system sprint begins.

8.3 Versioning and Tools

- Requirements Versioning — Requirements are version-controlled alongside the system's Software Requirements Specification (SRS); each release increments a major.minor version number (e.g., v1.0 → v1.1) to track software updates and dispatch logic refinements.
- Traceability Repository — A shared cloud spreadsheet (or lightweight requirements-management tool) hosts the live traceability matrix, accessible to the Change Control Board (Association Management, Technical Lead, and Dispatch Operations Head).
- Artifact Archival — All elicitation artifacts (e.g., passenger survey feedback, boda-boda rider focus group notes, stage representative workshop logs, control room observation records) are archived and referenced by ID from the requirements list for audit and historical tracking purposes.

9. Assumptions, Risks, and Sign-off

9.1 Assumptions

- All passengers and boda-boda riders have access to an active mobile money account (MTN MoMo or Airtel Money) or cash to settle ride fares.
- The mobile money providers' payment gateway APIs (MTN MoMo and Airtel Money) will remain stable and accessible throughout development and live operations.
- The low-cost cloud server infrastructure selected by the association will have sufficient operational capacity and bandwidth once optimized (per constraint R6).

9.2 Risks Identified During Elicitation

Risk	Likelihood / Impact	Mitigation
Mobile money network downtime (MTN MoMo / Airtel Money) during peak ride hours.	Medium / High	Support cash payments as a seamless fallback option and maintain encrypted offline transaction logs for delayed reconciliation.
Stakeholders disagree on fare calculation algorithms or stage commission policies after development starts.	Low / High	Formal sign-off on core fare structure, stage rules, and business logic before initiating UI/backend design (Section 9.3).
	Medium / High	Load-test against R10 before go-live; deploy auto-scaling cloud server resources to

Underestimated concurrent load and real-time GPS tracking traffic during Kampala evening rush hours.		dynamically absorb traffic spikes.

9.3 Sign-off

The requirements captured in this document, as refined in Section G.2, have been reviewed and are approved as the basis for producing the Software Requirements Specification and proceeding to system design.

Name	Role	Signature	Date
	Association Chairman / Management Representative		
	Head of Dispatch Operations		
	Technical Lead / Lead Developer		
	Project Manager / Business Analyst		